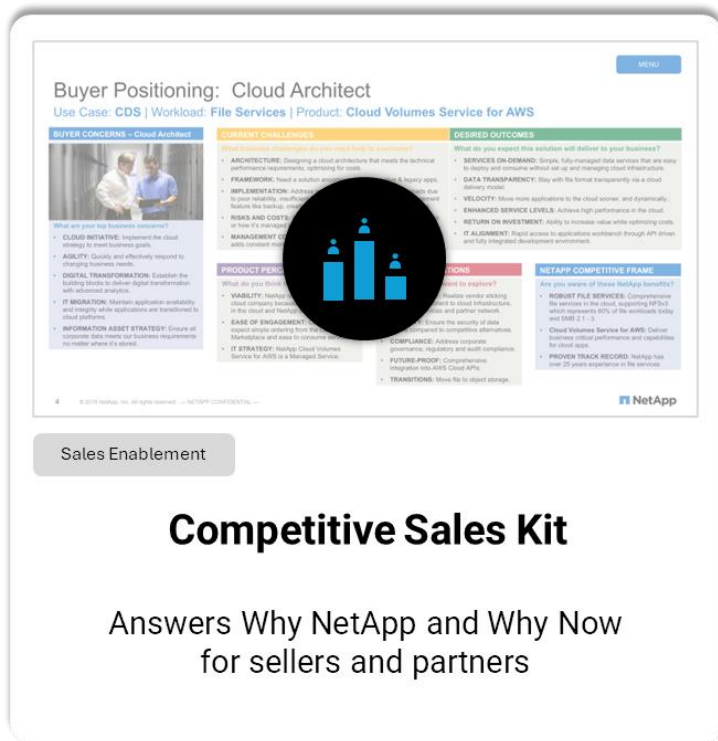




The Situation: NetApp was competing in cloud file services against AWS EFS, a service sold by a company that was simultaneously NetApp's most important strategic partner. AWS EFS had first-mover advantage, AWS marketplace integration, and a pricing model that looked simple but got expensive fast. NetApp's Cloud Volumes Service for AWS was newer, less familiar to buyers, and had to win on technical merit rather than brand default.

The Complication: The core problem was buyer perception, not product. Cloud architects defaulted to EFS because it was native AWS. They didn't know EFS was limited to NFSv4, capped at 7,000 IOPS and 16 IP connections per mount, and more expensive than NetApp at scale. NetApp supported NFSv3 (80% of actual file workloads), delivered up to 55,000 IOPS, and handled hundreds of connections. The sales team needed a way to expose those gaps without attacking a strategic partner.

The Resolution: We built a complete competitive sales kit that gave the field a framework for the AWS conversation and battle cards against HPE and SoftNAS. The kit reframed the partner tension as a complement story while equipping reps to attack EFS on IOPS ceilings, protocol gaps, copy inefficiency, and true cost at scale. It included a heat map for qualifying when to pursue, evaluate, or step aside, and buyer-specific positioning for cloud architects focused on migration risk, performance, and ROI.

[Scroll down to see the details](#)





Competitive Sales Kit

Use Case: **Cloud Data Services**

Workload: **File Services**

Product: **Cloud Volumes Service for AWS**

Strategic Partnerships

Although we may compete in certain market categories, Amazon is also an important strategic partner. Strategic alliance partners are collaborators, working alongside us to address today's most pressing business changes. They often tie directly into our portfolio teams, contributing their latest innovations and best practices to create, develop and deliver integrated solutions. They can also leverage their existing relationships to open new opportunities, influence companies to work with NetApp, and act as a reference to for us.

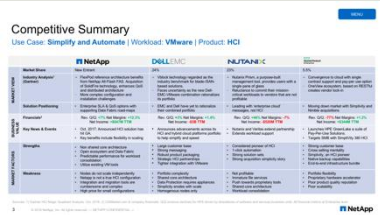
March, 2019



Menu

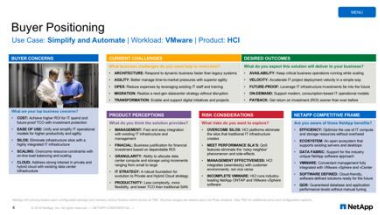
Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

Use the buttons to navigate this presentation.



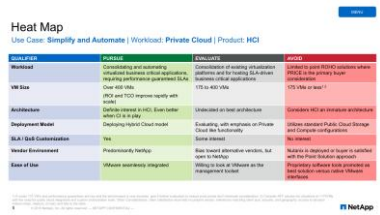
Competitive Summary
Includes Amazon Web Services (AWS), HPE, and SoftNAS

[VIEW NOW](#)



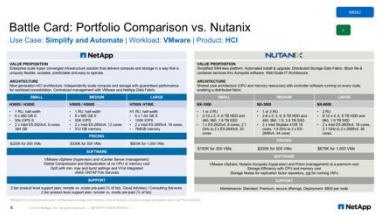
Buyer Positioning
Includes buyer profile, challenges, outcomes, and more

[VIEW NOW](#)



Heat Map
Includes guidelines for when to pursue, evaluate, and avoid

[VIEW NOW](#)



Battle Cards
Includes portfolio comparison, sales messaging, and technical messaging



[VIEW NOW](#)



[VIEW NOW](#)



[VIEW NOW](#)

Competitive Summary

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

		 NetApp			
MARKET VIEW	Solution Type	Cloud Service providing extreme performance, low latency and high IOPS.	Native Cloud Service (Amazon EFS)	Third-party Cloud Service	Storage Software Only Product with heavy touch configuration and mgt.
	Industry Analysis¹	+ Very robust cloud integration story, leading the industry. Data Fabric vision resonates very strongly with customers and analysts	+ Amazon Web Services (AWS) experiences rapid innovation driven by large user base. - Complex pricing models are difficult for customers to understand and predict.	+ Built around existing enterprise-grade storage products. - Integration of Nimble may negatively impact business and customers.	+ Early with NAS services in AWS and Azure. Early adopter customers. - Small vendor with limited resources facing pressure from larger vendors.
	Solution Positioning	- The industry's first multi-protocol file system service in the cloud. - Available as NetApp Cloud Volumes Service for AWS built on proven ONTAP technology and other hybrid cloud data services.	- Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage for use with AWS Cloud services and on-premises resources. - Amazon is a NetApp strategic partner.	- HPE Cloud Volumes provides an enterprise-grade multicloud block storage service for running applications in AWS and Azure. - Provides block storage only.	- Software-defined NAS for AWS and Azure and supporting NFS, SMB, iSCSI, and AFP file protocols. - A product customers configure to run in the cloud rather than a service.
BUSINESS VALUE	Financials²	Revenue: \$6B TTM Revenue TTM%: +2% Gross Margin: 65%	Revenue: \$23.3B TTM Revenue TTM % +9% Gross Margin: 25%	Revenue: 2.7B TTM Revenue TTM % +13 Gross Margin: 30.6%	Revenues: ~\$8M TTM Venture Capital Funding: \$26M Est Employees: 50-80 Est.
	Key News & Events	Purchase of Stackpoint.io to further support NetApp's main cloud initiatives and multi-cloud orchestration capabilities.	Announced support for encryption of data at rest, Jan. 2018.	Completes acquisition of Nimble in April 2017, then launches Cloud Volumes as HPE Cloud Volumes.	Announced new versions of Cloud Data Platform product: Oct. 2017.
MARKET FACTORS	Strengths	- First multicloud file storage service. - Integrated with the NetApp Data Fabric. - File storage service built for cloud accessible through cloud providers including Amazon. - Consumed directly from cloud marketplaces and billed by cloud provider.	- Complimentary portfolio of storage services including windows (Amazon FSx for Windows and Luster), block (EBS), object (S3), and archive (Glacier). - A global provider of compute, storage, database, migration, and networking, and content delivery cloud services.	- First multicloud block storage service. - Complimentary HPE cloud strategy, implementation, and support, and other consulting services.	- A software-defined storage (SDS) provider focused on rapid growth of cloud storage (file and block). - Customers include Netflix, Reuters, Boeing, and other major brands.
	Weakness		- Intense competition with Google and Microsoft for cloud services. - Regional Data center model creates both compliance and D/R concerns.	- Not consumable from marketplaces. - Technology from Nimble not HPE. - Sits apart from HPE storage portfolio raising roadmap concerns.	- Do-it-yourself (DIY) storage solution. - Reliant upon third-party cloud storage and compute infrastructure. - The company remains a start-up.

1) Analyst views aggregated by NetApp CI. Sources include, 451, Gartner, Forrester, IDC and Moore Insights & Strategy. 2) Sources: S.E.C. filings. HPE Storage revenues within the new HPE Hybrid IT Business unit. Amazon revenues reported at AWS B.U. level. SoftNas financials are estimates based upon secondary analyst reports and company press releases. TTM = Trailing Twelve Months Revenues

Buyer Positioning: Cloud Architect

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

BUYER CONCERNS – Cloud Architect



What are your top business concerns?

- **CLOUD INITIATIVE:** Implement the cloud strategy to meet business goals.
- **AGILITY:** Quickly and effectively respond to changing business needs.
- **DIGITAL TRANSFORMATION:** Establish the building blocks to deliver digital transformation with advanced analytics.
- **IT MIGRATION:** Maintain application availability and integrity while applications are transitioned to cloud platforms.
- **INFORMATION ASSET STRATEGY:** Ensure all corporate data meets our business requirements no matter where it's stored.

CURRENT CHALLENGES

What business challenges do you need help to overcome?

- **ARCHITECTURE:** Designing a cloud architecture that meets the technical performance requirements, optimizing for costs.
- **FRAMEWORK:** Need a solution encompassing cloud native & legacy apps.
- **IMPLEMENTATION:** Address migration delays of file-based workloads due to poor reliability, insufficient performance and the lack of data management feature like backup, creating copies and moving data between instances.
- **RISKS AND COSTS:** Changes to the application, refactoring native formats or how it's managed introduces risk and increase costs.
- **MANAGEMENT COMPLEXITY:** Implementing IaaS for custom file services adds constant monitoring and management complexity.

DESIRED OUTCOMES

What do you expect this solution will deliver to your business?

- **SERVICES ON-DEMAND:** Simple, fully-managed data services that are easy to deploy and consume without set up and managing cloud infrastructure.
- **DATA TRANSPARENCY:** Stay with file format transparently via a cloud delivery model.
- **VELOCITY:** Move more applications to the cloud sooner, and dynamically..
- **ENHANCED SERVICE LEVELS:** Achieve high performance in the cloud.
- **RETURN ON INVESTMENT:** Ability to increase value while optimizing costs.
- **IT ALIGNMENT:** Rapid access to applications workbench through API driven and fully integrated development environment.

PRODUCT PERCEPTIONS

What do you think the service provides?

- **VIABILITY:** NetApp is focused on being a cloud company because customers see value in the cloud and NetApp is customer driven.
- **EASE OF ENGAGEMENT:** Customers expect simple ordering from the Cloud Marketplace and ease to consume service.
- **IT STRATEGY:** NetApp Cloud Volumes Service for AWS is a Managed Service.

RISK CONSIDERATIONS

What risks do you want to explore?

- **VENDOR TRUST:** Realize vendor sticking power & commitment to cloud infrastructure, consulting services and partner network.
- **SECURITY:** Ensure the security of data assets compared to competitive alternatives.
- **COMPLIANCE:** Address corporate governance, regulatory and audit compliance.
- **FUTURE-PROOF:** Comprehensive integration into AWS Cloud APIs.
- **TRANSITIONS:** Move file to object storage.

NETAPP COMPETITIVE FRAME

Are you aware of these NetApp benefits?

- **ROBUST FILE SERVICES:** Comprehensive file services in the cloud, supporting NFSv3 which represents 80% of file workloads today and SMB 2.1 - 3.
- **Cloud Volumes Service for AWS:** Deliver business critical performance and capabilities for cloud apps.
- **PROVEN TRACK RECORD:** NetApp has over 25 years experience in file services.

Heat Map

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

QUALIFIER	PURSUE	EVALUATE	AVOID
Cloud Roadmap	Cloud architect with strategic direction to adopt cloud	Early stages of evaluation toward cloud	
Workloads	- HPC and Analytics: genomic sequencing, EDA, software builds, oil and gas 3D modeling, financial simulation models, fluid dynamics, Hadoop, machine learning and IoT - Files services: home directories, software development, online content, and web serving - Windows SMB workloads: SQL, File Shares	- Backup, disaster recovery, and DevOps - Consider engaging NetApp Advisory services to help guide your customer	
Vendor Preference	- NetApp customer - Amazon EFS customer (faster snapshots and copies) - Isilon customer (Isilon has no files to cloud solution)	Not a NetApp customer, but open to Data Fabric	
Interested in File Services in the Cloud	- Customer has mandate to move files to the cloud - Customer would move NFSv3 to the cloud - Customer would move SMB 2.1 - 3 to the cloud	- Customer interested in moving files to the cloud - Customer has moved NFSv4 to the cloud - Customer want to move databases to the cloud (engage NetApp databases team)	
Interested in Blocks Services in the Cloud 1	Present Cloud Volumes ONTAP instead of Cloud Volumes Service for AWS	Review if block need is preference or requirement. Many workloads run with files.	Customer needs block support (use Cloud Volumes ONTAP)
Customization	Customer would use NetApp Cloud Volumes Service for AWS without expecting feature changes	Customer expects capabilities beyond NetApp Cloud Volumes Service for AWS (consider Cloud Volumes ONTAP)	Customer needs customization or block storage (use Cloud Volumes ONTAP)



Battle Card: NetApp vs. AWS EFS

Use Case: **Cloud Data Services**

Workload: **File Services**

Product: **Cloud Volumes Service for AWS**

Strategic Partnerships

Although we may compete in certain market categories Amazon-AWS is also an important strategic partner. Strategic alliance partners are collaborators, working alongside us to address today's most pressing business changes. They often tie directly into our portfolio teams, contributing their latest innovations and best practices to create, develop and deliver integrated solutions. They can also leverage their existing relationships to open new opportunities, influence companies to work with NetApp, and act as a reference to for us.

Battle Card: Portfolio Comparison vs. AWS EFS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**



VALUE PROPOSITION

NetApp Cloud Volumes Service for AWS is a fully-managed file storage service that's easy to consume and use. It delivers unprecedented performance to your cloud applications and development environments. Cloud Volumes Service for AWS offers advanced data management capabilities, so you don't have to build data management into your environment, and you can spend your time on more important activities.

ARCHITECTURE

Comprehensive cloud file service purchased through the hyperscaler, offering multiprotocol file services to applications in the cloud.

SMALL	MEDIUM	LARGE
20 TB	50 TB	100TB
Fully managed service - NFSv3, NFSv4, and SMB - Rapid copies - Rapid snapshots - Snapshot scheduling	Fully managed service - NFSv3, NFSv4, and SMB - Rapid copies - Rapid snapshots - Snapshot scheduling	Fully managed service - NFSv3, NFSv4, and SMB - Rapid copies - Rapid snapshots - Snapshot scheduling

PRICING

\$60,000 per year	\$150,000 per year	\$300,000 per year
• Pay-as-you-go planned	• Pay-as-you-go planned	• Pay-as-you-go planned

SOFTWARE

Includes Cloud Sync (no additional fee)

SUPPORT

Provided by NetApp



VALUE PROPOSITION

Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage for use with AWS Cloud services and on-premises resources. It is easy to use and offers a simple interface that allows you to create and configure file systems quickly and easily. Amazon EFS is built to elastically scale on demand without disrupting applications, growing and shrinking automatically as you add and remove files, so your applications have the storage they need, when they need it.

ARCHITECTURE

AWS EFS is a software instantiated NFSv4 platform that is consumption based without minimum purchase requirements.

SMALL	MEDIUM	LARGE
20 TB	50 TB	100TB
Fully managed service - NFSv4 only - Standard, full copies - Data copy only	Fully managed service - NFSv4 only - Standard, full copies - Data copy only	Fully managed service - NFSv4 only - Standard, full copies - Data copy only

PRICING

>\$71,349 per year	>\$178,374 per year	>\$365,748 per year
• \$0.30+ per GB per month	• \$0.30+ per GB per month	• \$0.30+ per GB per month

SOFTWARE

Offers Amazon FSx for Windows at a high price range \$.17/GB, 64TB max volume size, \$2GB/s aggregate max
Offers EFS File Sync and AWS DataSync for additional fee

SUPPORT

Provided by Amazon AWS

Battle Card: Sales Messaging vs. AWS EFS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**



aws	NetApp
WEAKNESSES	ATTACK
IOPS	- AWS EFS has a limit of 7,000 IOPS max and 250MiB/s max per mount. - Cloud Volumes Service for AWS scales up to ~55,000 IOPS and up to 1024MiBps.
Data Copies	- AWS EFS requires a full copy of everything, every time. - NetApp provides efficient instant snapshot copies, so it's more cost-efficient. - NetApp copies datasets, AWS does not.
File Services Support	- NetApp supports NFSv3 and SMB (NFSv4 is coming soon. (AWS only supports NSFv4). - NFSv3 accounts for ~80% of all data.
Scale (Fan in)	- If the customer needs a lot of users/applications to connect to a share, EFS limits only 16 IP addresses that can communicate with a volume. - NetApp can scale far beyond hundreds of IP host addresses.
Performance	- EFS is targeted for low IOPS application requirements or home directories. - NetApp Cloud Volumes Service for AWS supports up to 20X the performance of EFS with up to 4000 IOPS/TB.

aws	NetApp
STRENGTHS	FLANK OR COUNTER
Scale (size)	- AWS EFS can grow to PB scale. - NetApp supports 100TB volumes with road maps to petabyte scale
Flexibility	- AWS EFS is mountable from multiple Availability Zones within a VPC. - Connect to multiple VPCs via Direct Connect link. Cloud Volumes Service for AWS can also support multiple availability zones within a VPC.
Encryption	- EFS supports data encryption is supported at rest and in flight. - NetApp provides at rest encryption and recommends 3rd party network based encryption and in-flight with SMB 3 and NFSv4.1 once available.
Dominant Player in Cloud	- AWS offers a robust set of complimentary services with rapid release cycles. It's - NetApp is focused on storage, leveraging deep NFS and SMB expertise.
Elasticity	- AWS has a broad range of options. - NetApp offers multiple subscription models with a short-term roadmap to introduce flexible consumption options.
Global Coverage	- EFS is limited to specific geographic cloud data centers. - NetApp is available in the Americas and Europe and has a roadmap to expand to global coverage. - NetApp has true global coverage via ONTAP Cloud in all AWS regions .

aws	NetApp
COMMON FUD	INOCULATE YOUR CLIENT
EFS is a De Facto Standard	- A cloud service is often very attractive. - EFS fits well for only a limited set of file services. - Cloud Volumes Service for AWS brings enhanced data management functionality and unprecedented performance
Complimentary Block and Object Services	- Separate service offerings required for each service. - NetApp offers Cloud Volumes ONTAP for universal block and file support. - NetApp has additional object solutions for build your own environment.
Oracle database	- Amazon RDS is used for Oracle databases. - Cloud Volumes Service for AWS runs Oracle databases, offers performance and data management e.g. snapshots, clones.
DIY NAS v. Fully Managed Service	- DIY NAS requires a lot of manual configuration and data management capabilities vary. - Cloud Volumes Service for AWS offers shared storage, scale, performance, snapshots, cloning and other data management as a fully managed service.

Battle Card: Technical Messaging vs. AWS EFS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**



KEY MESSAGE	HOW IT IS ACHIEVED	NETAPP ADVANTAGE
Strategic Partnership	Close development work with AWS	NetApp adds to the capabilities of AWS. When competing, we must highlight NetApp value and not attack AWS.
Business Critical Capabilities	Storage availability and durability	ONTAP provides world class availability and durability characteristics, witnessed by the 100's of thousands of ONTAP customers for over 25 years.
	Predictable performance	- Cloud Volumes Service for AWS are intelligently managed with QoS and offers predictable performance - Cloud Volumes Service for AWS do not have artificial limits, therefore can deliver extreme file performance to applications.
	Advanced data management	NetApp zero-impact Snapshot copies, Snapshot scheduling, rapid copies, restore from copies.
Hybrid Cloud Mobility	Cloud Sync, integrated data movement service for migration	- Transfer data to or from Cloud Volumes Service for AWS seamlessly and securely dropping transfer time from hours to minutes, saving both time and money. Cloud Sync is included in the Cloud Volumes Service for AWS. - AWS offers EFS File Sync, which is faster than host copies, but does track changed blocks, limiting efficiency and AWS DataSync. Both are separate solutions and charged at a different price range.
Enhanced Cloud Economics	Rest API	All UI actions are fully automatable using REST API for programmatic integration.
No technical tradeoffs	No fan-in limitations	NetApp supports hundreds of IP addresses vs. AWS limited to 16.
	NFSv3 and NFSv4 and SMB 3.0	- Cloud Volumes Service for AWS runs NFSv3, NFSv4 (soon) and SMB in the same fully managed service. - AWS is NFSv4 only and offers SMB through a separate service called Amazon FSx under a different pricing structure.



Battle Card: NetApp vs. HPE Cloud Volumes

Use Case: **Cloud Data Services**

Workload: **File Services**

Product: **Cloud Volumes Service for AWS**

Battle Card: Portfolio Comparison vs. HPE Cloud Volumes

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

NOTE: HPE Cloud Volumes is not a direct competitor to NetApp Cloud Volumes Service for AWS. HPE Cloud Volumes cannot be purchased through the AWS marketplace and is block only.



VALUE PROPOSITION

NetApp Cloud Volumes Service for AWS is a fully-managed file storage service an easy to consume and use. It delivers unprecedented performance to your cloud applications and development environments. Cloud Volumes Service for AWS offers advanced data management capabilities, so you don't have to build data management into your environment, and you can spend your time on more important activities.

ARCHITECTURE

Comprehensive cloud file service purchased through the hyperscaler, offering multiprotocol file services to applications in the cloud.

SMALL	MEDIUM	LARGE
20 TB	50 TB	100TB
Fully managed service • NFSv3, NFSv4, and SMB • Rapid copies • Rapid snapshots • Snapshot scheduling	Fully managed service • NFSv3, NFSv4, and SMB • Rapid copies • Rapid snapshots • Snapshot scheduling	Fully managed service • NFSv3, NFSv4, and SMB • Rapid copies • Rapid snapshots • Snapshot scheduling
PRICING		
\$60,000 per year • Pay-as-you-go planned	\$150,000 per year • Pay-as-you-go planned	\$300,000 per year • Pay-as-you-go planned
SOFTWARE		
Includes Cloud Sync (no additional fee)		
SUPPORT		
Included, provided by NetApp		



VALUE PROPOSITION

HPE Cloud Volumes offers enterprise-grade multicloud storage service for Amazon Web Services (AWS) and Microsoft Azure. It is a a block storage service that delivers enterprise-grade reliability and security with data protection and copy data management features needed by enterprise applications.

ARCHITECTURE

Cloud consumption service offering **block storage services** to applications in the cloud. Theoretically, can be used to build DIY file servers in the cloud using using third-party software running on compute instances.

SMALL	MEDIUM	LARGE
20 TB	50 TB	100TB
Fully managed service • iSCSI block only • Optional cloning, snapshots, replication • Paid additional IOPS	Fully managed service • iSCSI block only • Optional cloning, snapshots, replication • Paid additional IOPS	Fully managed service • iSCSI block only • Optional cloning, snapshots, replication • Paid additional IOPS
PRICING		
>\$24,000 per year • Storage: \$0.10+ GB/month	>\$60,000 per year • Storage: \$0.10+ GB/month	>\$120,000 per year • Storage: \$0.10+ GB/month
SOFTWARE		
Cloning, snapshots, replication (additional fee for each)		
SUPPORT		
12% add-on for unlimited 24x7 technical support (phone/email), provided by HPE		

Battle Card: Technical Messaging vs. HPE Cloud Volumes

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

NOTE: HPE Cloud Volumes is not a direct competitor to NetApp Cloud Volumes Service for AWS.
HPE Cloud Volumes cannot be purchased through the AWS Marketplace and is block only.

KEY MESSAGE	HOW IT IS ACHIEVED	NETAPP ADVANTAGE
File Services	Multiprotocol file access	- NetApp Cloud Volumes Service for AWS offers direct file service supporting NFSv4, NFSv3, and SMB 3.0 protocols - To support file services, HPE Cloud Volumes would require the customer to build a custom file environment on top of the HPE block service at a considerable additional cost with serious performance and manageability limitations - For customers requiring cloud storage services, NetApp offers ONTAP Cloud
	Cloud file service built for cloud	- NetApp Cloud Volumes Service for AWS brings the value of file storage and access to applications - Benefits of files storage include better access rights, no LUN limitations, and simplified management
Hybrid Cloud Mobility	Cloud Sync, integrated data movement service for migration	- Transfers data between NetApp Cloud Volumes Service for AWS seamlessly and securely while reducing transfer time from hours to minutes to save time and money - Transfer data between heterogeneous file environments (file servers, NAS systems, on-premise and cloud)
No technical tradeoffs	No fan-in limitations for storage access	NetApp Cloud Volumes Service AWS can support hundreds of compute instances simultaneously unlike HPE
Cloud-integrated service	Complete integration with AWS cloud	- NetApp Cloud Volumes Service for AWS is available for purchase from AWS Marketplace - Purchase and pay from within your AWS bill - NetApp is a highly integrated cloud service that does not require storage domain expertise

Battle Card: Sales Messaging vs. HPE Cloud Volumes

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

NOTE: HPE Cloud Volumes is not a direct competitor to NetApp Cloud Volumes Service for AWS. HPE Cloud Volumes cannot be purchased through the AWS marketplace and is block only.

Hewlett Packard Enterprise	
NetApp	
WEAKNESSES	ATTACK
Block Only	- HPE Cloud Volumes is an iSCSI block storage service. - HPE file services would need to be built at a considerable cost and limiting performance and manageability. - NetApp Cloud Volumes Service for AWS is a file service offering with high performance and simple management. - If block service is required, consider Cloud Volumes ONTAP.
Hyperscaler Integration	- HPE Cloud Volumes is a third party cloud service with separate billing by HPE. - NetApp Cloud Volumes Service for AWS is fully integrated with the hyperscaler to simplify deployment, operation, and billing. - Purchase NetApp Cloud Volumes Service for AWS directly from the AWS Marketplace.
Hybrid Operation	- HPE Cloud Volumes only integrates with customers using Nimble storage. - NetApp Cloud Volumes Service for AWS integrates with all ONTAP systems. - NetApp Cloud Volumes Service for AWS migrates data from any file environment using the integrated NetApp Cloud Sync service

Hewlett Packard Enterprise	
NetApp	
STRENGTHS	FLANK OR COUNTER
Enterprise-class	- HPE Cloud Volumes bring the power of enterprise capabilities to a cloud storage service. - NetApp Cloud Volumes Service for AWS brings the power of ONTAP to the cloud.
Multi-cloud Support	- HPE Cloud Volumes can be deployed from the service and connect to a choice of hyperscalers. - NetApp Cloud Volumes Service for AWS is integrated with the hyperscaler, but solutions are available from each of the major providers offering the value integrated from each.
Hybrid Cloud Support	- HPE Cloud Volumes address the cloud but work with cloud compute but can integrate with on-premises HPE storage. - HPE on premise integration only works with Nimble based systems. - NetApp Cloud Volumes Service for AWS works with all ONTAP on-premises environments and heterogeneous environments with the Cloud Sync service.

Hewlett Packard Enterprise	
NetApp	
COMMON FUD	INOCULATE YOUR CLIENT
HPE Cloud Volumes is a True Cloud Service	HPE is a 3rd party service that connects to the cloud and is not an integrated offering in the AWS Marketplace. It is not integrated into the hyperscaler cloud like NetApp Cloud Volumes Service for AWS.



Battle Card: NetApp vs. SoftNAS

Use Case: **Cloud Data Services**

Workload: **File Services**

Product: **Cloud Volumes Service for AWS**

Battle Card: Portfolio Comparison vs. SoftNAS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

NetApp

VALUE PROPOSITION

NetApp Cloud Volumes Service for AWS is a fully-managed file storage service an easy to consume and use. It delivers unprecedented performance to your cloud applications and development environments. Cloud Volumes Service for AWS offers advanced data management capabilities, so you don't have to build data management into your environment, and you can spend your time on more important activities.

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PRICING		
\$60,000 per year Pay-as-you-go planned	\$150,000 per year Pay-as-you-go planned	\$300,000 per year Pay-as-you-go planned
SOFTWARE		
Includes Cloud Sync (no additional fee)		
SUPPORT		
Included, provided by NetApp		

SoftNAS

VALUE PROPOSITION

SoftNAS makes it easy for businesses to tap into the full potential of cloud data storage without the effort and expense required of “application modernization,” while transitioning from on-premises storage to the public or hybrid cloud, or even to multi-cloud environments.

ARCHITECTURE

SoftNAS Cloud Enterprise is a software-defined, full-featured enterprise cloud NAS filer products for primary data storage that runs on AWS and Azure to deliver block and file storage.

SMALL	MEDIUM	LARGE
20 TB	50 TB	100TB
Custom software-defined storage - SoftNAS Enterprise HP/HA - AWS M4.2xlarge compute - EBS storage (gp2)	Custom software-defined storage - SoftNAS Enterprise HP/HA - AWS M4.2xlarge compute - EBS storage (gp2)	Custom software-defined storage - SoftNAS Enterprise HP/HA - AWS M4.2xlarge compute - EBS storage (gp2)
PRICING ¹		
>\$53,000 per year - SoftNAS: \$21,882 per year (bundle with 30% off) - EC2: \$7,008 per year - EBS: \$24,000 per year	>\$121,000 per year - SoftNAS: \$54,706 per year (bundle with 30% off) - EC2: \$7,008 per year - EBS: \$60,000 per year	>\$236,000 per year (est.) - SoftNAS: \$109K per year (bundle with 30% off) - EC2: \$7,008 per year - EBS: \$120,000 per year
SOFTWARE		
SoftNAS is a software product managed in the Cloud by the customer		
SUPPORT		
SoftNAS 24x7x365 Gold-level and AWS Basic support included		

1) SoftNAS 20TB and 50TB pricing is bundled annual pricing from AWS marketplace.

Battle Card: Sales Messaging vs. SoftNAS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**

 SoftNAS	 NetApp
WEAKNESSES	ATTACK
Not a Cloud Service	Software product requiring high touch customization and operations.
Manageability	- Customer has to manage the SoftNAS software, patch it, maintain it, etc. - No automated deployment of a SoftNAS solution, very complex and time consuming.
High Availability	- For high availability and DR needs, our NetApp Cloud Volumes ONTAP High Availability solution in AWS ensures no data loss in the event of an outage of one of the nodes. SoftNAS Cloud Cannot guarantee the same. - High Availability configurations in SoftNAS require additional licenses and configuration resources.
Support	Support for a SoftNAS Cloud system requires a separate annual support contract.

 SoftNAS	 NetApp
STRENGTHS	FLANK OR COUNTER
Flexibility	- Supports NFSv3, iSCSI and SMB. - Object Storage Support. - AWS Simple Storage Service (S3) support. - Azure Blob Storage support. - Customers like because of cheaper storage costs. - Azure storage support, HA in Azure.
Appeals to Storage Administrators	- NetApp offers a service versus product, enabling users to spend more time leveraging their data, reducing specialized requirements and speeding time to deploy. - SoftNAS configurations and maintenance requires dedicated personnel.
Supports iSCSI Block	Counter with Cloud Volumes ONTAP that supports iSCSI.
No Down Time Guarantee	Service level with limited coverage and downtimes.

 SoftNAS	 NetApp
COMMON FUD	INOCULATE YOUR CLIENT
SoftNAS was First to Market with NAS Services in AWS and Azure	- A cloud product is often very attractive. SoftNAS forces significant tradeoffs and added management investment. - Cloud Volumes Service for AWS brings dramatic simplicity without system management requirements.
SoftNAS Offers Native Object Storage Support	- Ask the customer why they're interested in Object Storage. - For customers requiring Object storage NetApp offers StorageGRID® Webscale. Store and manage unstructured data at scale using StorageGRID Webscale for secure, durable object storage. Take control of rich content and transform your business.

Battle Card: Technical Messaging vs. SoftNAS

Use Case: **CDS** | Workload: **File Services** | Product: **Cloud Volumes Service for AWS**



KEY MESSAGE	HOW IT IS ACHIEVED	NETAPP ADVANTAGE
Business Critical Grade Capabilities	Storage availability and durability	- NetApp Cloud Volumes Service for AWS provides world class availability and durability characteristics. - NetApp Cloud Volumes Service for AWS leverages the 100's of thousands of customer experience for over 25 years.
	Predictable performance	- NetApp Cloud Volumes Service for AWS are intelligently managed with QoS and offers predictable performance. - SoftNAS performance is heavily dependent upon architecture and configuration which the customer must choose.
	Advanced data management	NetApp zero-impact Snapshot copies, Snapshot scheduling, rapid copies, restore from copies.
Hybrid Cloud Mobility	Cloud Sync, a data movement tool for migration to cloud volumes	- Transfers data to/from NetApp Cloud Volumes Service for AWS seamlessly and securely dropping transfer time from hours to minutes, saving both time and money. - Transfer data to and from heterogeneous File environments (On-Prem. NAS vendors, Other Cloud environments).
Enhanced Cloud Economics	Rest API	- UI actions are fully automatable. - UI is fully REST API ready for programmatic integration.
No technical tradeoffs	No Fan-in limitations	In AWS, limited to 16 IP addresses. NetApp supports hundreds of IP addresses.
	Performance limited to speed of the underlying Hyper Scale Cloud resources	- NetApp offers independent protected Cloud resources so that performance is not impacted by hyperscaler resource limitations. - Multi-tenant and QoS, delivering workload isolation.
Managed Service	NetApp Cloud Volumes Service for AWS is a complete, fully-managed cloud service	- NetApp is available and ready-to-use in AWS Marketplace whereas SoftNAS is a do-it-yourself software-defined product. - NetApp is a complete service which does not require storage domain expertise.



Thank You